Assignment - 3

Explain IaaS in brief.

IaaS - Infrastructure as a Service is a cloud service model that provides virtualized computing resources such as virtual machines, servers, storage and networking over the internet. Users can create and manage these resources according to their requirements without purchasing physical hardware.

: Amazon EC2, MS Azure VMs, Google Compute Engine.

Advantages:

- Reduces hardware and maintenance costs.
- Resources can be scaled up or down easily.
- Provides flexibility and on-demand resource allocation.

Discuss Virtual Provisioning in Virtualized Data Centre.

Virtual Provisioning is the process of dynamically allocating virtual computing resources such as CPU, memory, storage and network resources to virtual machines according to their requirements.

In a virtualized data centre, resources are managed by virtualization software or a hypervisor. Resources can be allocated, modified or released without changing the physical hardware.

Benefits:

- Better utilization of physical resources.
- Faster deployment of virtual machines.

Easy scalability and resource management.
Reduces operational and infrastructure code.

. Explain Virtual Machine Migration Services.
VM migration is the process of moving a virtual machine for one physical server or host to another. It can be performed while the VM is running, known as a live migration or when it is stopped.

VM Migration is used for load balancing, server maintenance, fault management and efficient resource utilization.

Types:
- Live Migration: VM is moved with little or no service interruption.

Offline Migration: VM is stopped befor being moved.

Benefits:
- Improved availability, load balancing, maintenance without major downtime and better resource utilization.

. Discuss Integration of Public and Private Cloud.
> The integration of Public and Private Cloud means connecting an organization's private cloud infrastructure with resources and services from a public cloud. This combination is commonly called as hybrid cloud.

In a hybrid cloud, sensitive data and critical applications can remain in the private cloud, while the public cloud can be used for additional computing resources, storage or handling increased workloads.

Advantages:
- Provides flexibility and scalability.
- Improves resource utilization.
- Keeps sensitive data under greater organizational control.
- Reduces infrastructure costs.
- Supports workload migration and backup.

Explain any three Tools and Technologies used in Cloud Computing.

1) Virtualization:
- Virtualization creates virtual versions of servers, storage, networks, or operating systems. It allows multiple virtual machines to run on a single physical machine.
- VMware and Hyper-V are common examples.

2) Containerization:
- Packages an application along with its dependencies so that it can run consistently across different environments.
- Docker is a widely used container technology

3) Cloud Management Tools:
- These tools help organizations monitor, configure, automate and manage cloud resources. They provide features such as resource monitoring, deployment, security and cost management.
- Examples are Openstack and Kubernetes.